

Randomized phase II trial of lyophilized strawberries in patients with dysplastic precancerous lesions of the esophagus.



Dysplasia is a histologic precursor of esophageal squamous cell carcinoma (SCC). We previously showed that dietary freeze-dried, or lyophilized, strawberry powder inhibits N-nitrosomethylbenzylamine-induced SCC in the rat esophagus. On the basis of this observation, we conducted a randomized (noncomparative) phase II trial in China to investigate the effects of two doses of freeze-dried strawberries in patients with esophageal dysplastic lesions in a high-risk area for esophageal cancer. We randomly assigned 75 patients identified by endoscopy to have dysplastic esophageal premalignant lesions to receive freeze-dried strawberry powder at either 30 g/d (37 patients) or 60 g/d (38 patients) for six months; the powder was mixed with water and drunk. After six months, we assessed the changes in histologic grade of these lesions (primary endpoint) in a blinded fashion. The dose of 30 g/d, did not significantly affect histology or any other measured parameter. The dose of 60 g/d, however, reduced the histologic grade of dysplastic premalignant lesions in 29 (80.6%) of the 36 patients at this dose who were evaluated for histology ($P < 0.0001$). The strawberry powder was well tolerated, with no toxic effects or serious adverse events. Strawberries (60 g/d) also reduced protein expression levels of inducible nitric oxide synthase (iNOS) by 79.5% ($P < 0.001$), cyclooxygenase-2 (COX-2) by 62.9% ($P < 0.001$), phospho-nuclear factor kappa B (NF κ B)-p65 (pNF κ B-p65) by 62.6% ($P < 0.001$), and phospho-S6 (pS6) by 73.2% ($P < 0.001$). Freeze-dried strawberries (60 g/d) also significantly inhibited the Ki-67 labeling index by 37.9% ($P = 0.023$). Our present results indicate the potential of freeze-dried strawberry powder for preventing human esophageal cancer, supporting further clinical testing of this natural agent in this setting. ©2011 AACR.